## Exercise 1 - Co-occurrence Matrix

Write a co-occurrence matrix for the following sentence:

> "A bird in the hand is worth two in the bush."

Count each word only if it appears directly after the reference word. Is the co-occurrence matrix unique?

## Exercise 2 - Convolutional Layers

Consider the following $4 \times 4 \times 1$ input X and a $2 \times 2 \times 1$ convolutional kernel K with no bias term

$$X = \begin{pmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -3 & 4 & 0 & 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 2, & 1 \\ 0, & 1 \end{pmatrix}$$

(a) What is the output of the convolutional layer for the case of stride 1 and no padding?

(b) What if we have stride 2 and no padding?

(c) What if we have stride 2 and zero-padding of size 1?

## Exercise 3 - Sizes in MLPs Refresher

You are given an MLP with ReLU activations. It has 3 layers consisting of 5, 10, and 5 neurons respectively. The input is a vector of size 10. How many parameters does this network have?

## Exercise 4 - Sizes in CNNs

You are givne a neural network with the following architecture:

```
Input: 100 x 100 x 3 Image

Layers:
1. Conv(in_channels=3, out_channels=5, kernel_size=3, stride=1, padding=0)
2. MaxPool2d(kernel_size=2, stride=2, padding=0)
3. Conv(in_channels=5, out_channels=10, kernel_size=3, stride=1, padding=0)
4. MaxPool2d(kernel_size=2, stride=2, padding=0)
5. Conv(in_channels=10, out_channels=5, kernel_size=3, stride=1, padding=0)
6. Flatten()
7. MLP(neurons=20)
8. MLP(neurons=10)
```

(a) What is the dimensionality of the activations after each layer.

(b) How many parameters does this network have?